

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 02



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Recap of Previous Lecture



Topic

Calendar

Date^u → Day?



Topics to be Covered



Topic-1

Clock

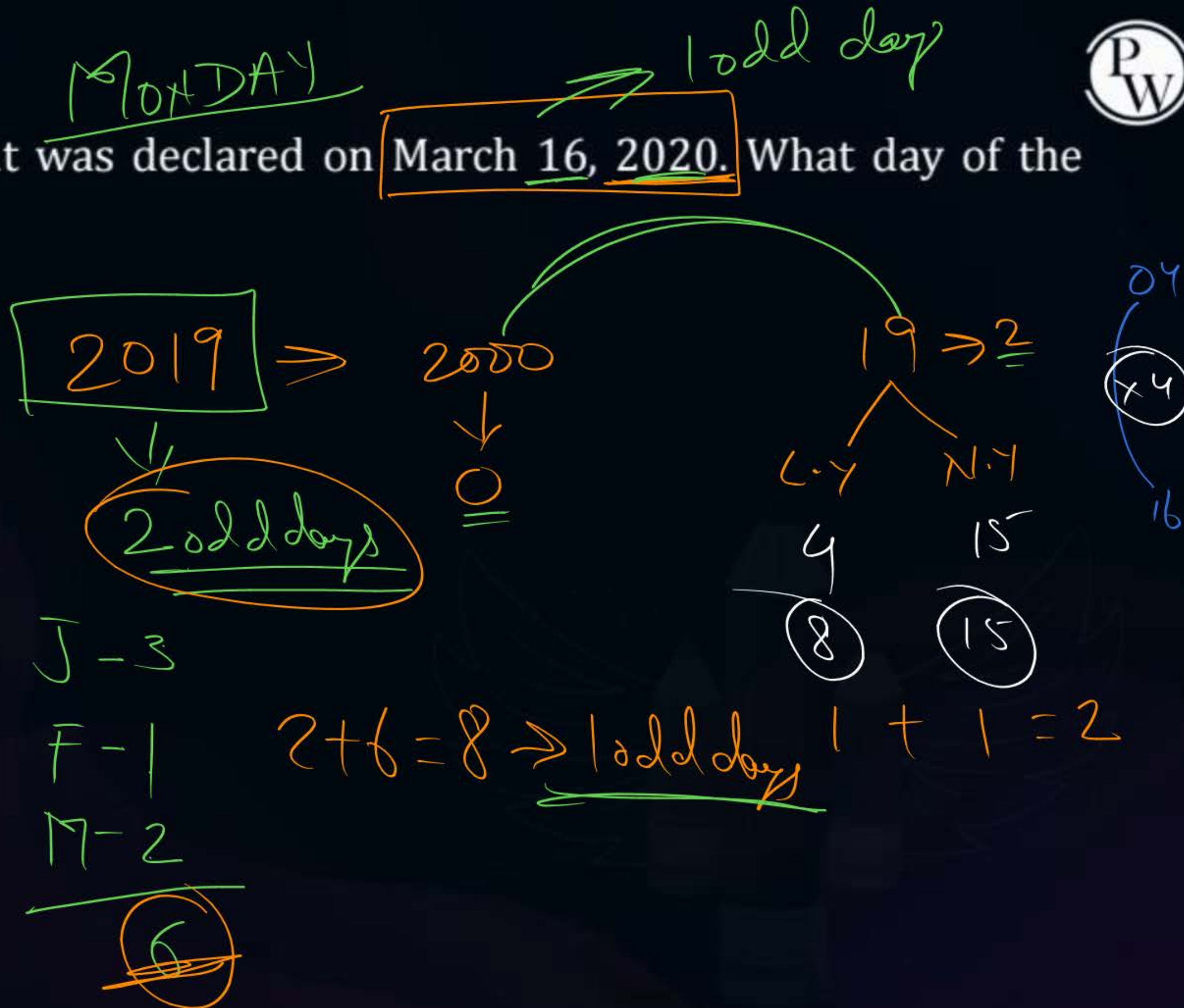


[MCQ]



#Q. The GATE 2020 result was declared on March 16, 2020. What day of the week was it?

- A** Saturday
- B** Sunday
- C** Monday
- D** None



[MCQ]



#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

Assignment

- A** Monday
- B** Thursday
- C** Tuesday
- D** Saturday

SATURDAY 6 odd days

1800 $\Rightarrow$ 3 $\checkmark$

68 $\Rightarrow$ 1 $\checkmark$

1868

4 odd days

J-~~3~~ x M-~~3~~ S-~~2~~ 17 34 6 + 2 = 8 (1 odd day)

F-0 J-~~2~~ 0-~~2~~ 51 51

M-~~3~~ J-~~3~~ A-~~3~~

1-2

04

X 17

68

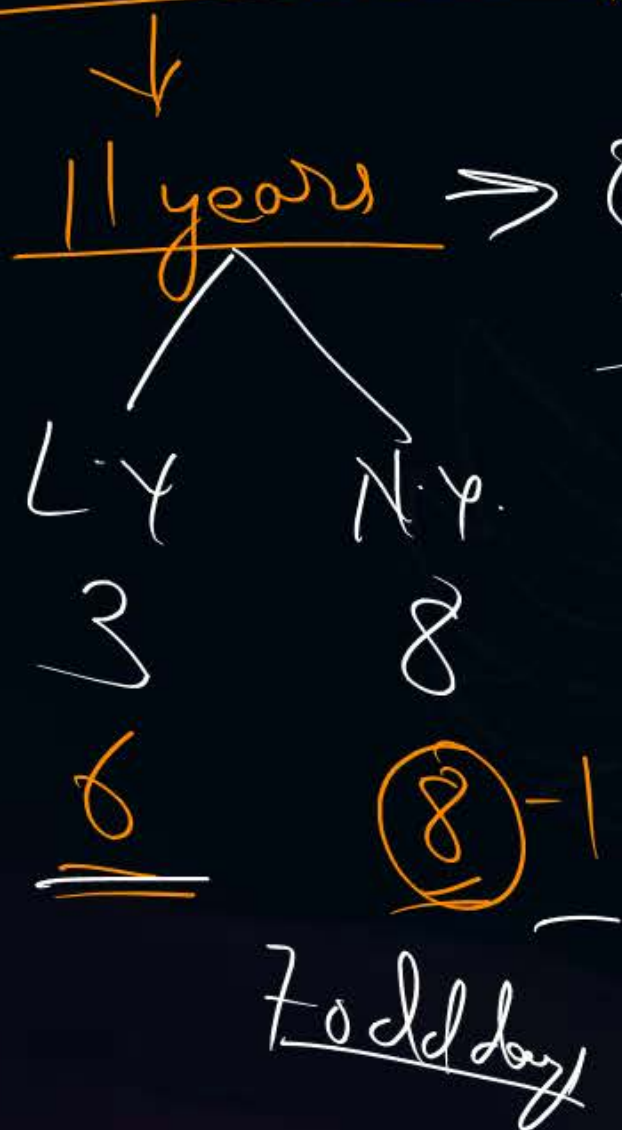
[MCQ]



#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

- A** Sunday
- B** Wednesday
- C** Thursday
- D** Friday

1992 - 2003



Assignment

✓	92
✓	96
✓	2000

+0 = Wednesday

#Q. A person was born on fifth Monday of February in a particular year. 29 leap
Which one of the following statements is correct based on the above information?

Assignment

A

The 2nd February of that year is a Tuesday. ✓

B

There will be five ~~Sundays~~ ^{five Mondays} in the month of February in that year. ✗

C

The 1st February of that year is a ~~Sunday~~ ^{Monday}. ✗

D

All Mondays of February in that year have even dates. ✗

2011

Cover the books

throwing

cleaning paper

Selling

CALENDAR REPETITION

2011 → 1

2012 → 2✓

2013 → 1✓

2014 → 1✓

2015 → 1✓

2016 → 2

2017 → 1

2018 → 1

2019 → 1

2020 → 2

2021 → 1

2022

[MCQ]



#Q. 2005 calendar is same as which of the given years?

A 2009

B 2016

C 2011

D 2015

2005 → 1
2006 → 1
2007 → 1
2008 → 2
2009 → 1
2010 → 1
2011

[MCQ]



#Q. When does 2016 calendar repeats?

- A** 2027
- B** 2040
- C** 2021
- D** 2044

L.Y.

$$\begin{array}{r} 2016 \\ + 28 \\ \hline 2044 \end{array}$$

$$\begin{array}{r} 2004 \\ + 28 \\ \hline 2032 \end{array}$$

$$\begin{array}{r} 2012 \\ + 28 \\ \hline 2040 \end{array}$$

L.Y. $\rightarrow$ 28 yrs
 $\rightarrow$ N.Y. $\rightarrow$ 6 years
 $\rightarrow$ N.Y. $\rightarrow$ 11 yrs
 $\rightarrow$ N.Y. $\rightarrow$ 11 yrs

L.Y. $\rightarrow$ 28 years

$$\begin{array}{r} 2017 \\ + 6 \\ \hline 2023 \end{array}$$

$$\begin{array}{r} 2005 \\ + 6 \\ \hline 2011 \end{array}$$

Conditional Question

Q. If a given year has 389 days and every week is of 9 days, then how many odd days that year has?

$$\begin{array}{r} 9 \overline{) 389} \quad (43 \\ \underline{36} \\ 29 \\ \underline{27} \\ 2 \end{array}$$

2 odd days

Conditional Question



Tue | Thu | Sat

Q The last day of any century CANNOT be which day?

100 - 5	→	<u>FRIDAY</u> ✓
200 - 3	→	<u>WEDNESDAY</u> ✓
300 - 1	→	<u>MONDAY</u> ✓
400 - 0	→	<u>SUNDAY</u> ✓

[MCQ]



#Q. The day of the March 16th of any year is the same day of the week as the corresponding date in which month of the same year?

- A** July
- B** November
- C** April
- D** September

March $\rightarrow 3$ Aug $\rightarrow 3$
April $\rightarrow 2$ Sep $\rightarrow 2$
May $\rightarrow 3$ Oct $\rightarrow 3$
June $\rightarrow 2$
July $\rightarrow 3$

Nov

#Q. For the year 2019, which of the previous year's calendar can be used?

A 2011

B ~~2012~~

C 2013

D 2014

2019 → 1

2018 → 1

2017 → 1

2016 → 2

2015 → 1

2014 → 1

2013

CLOCK



Relative Speed



Relative Speed:

A ~ B

60 kmph

80 kmph



opposite $\rightarrow$ (+)

Same $\rightarrow$ (-)

Relative Speed:

1. Hands

2. Movement

3. Relative Speed

Minimum Hands?

Same

Opposite

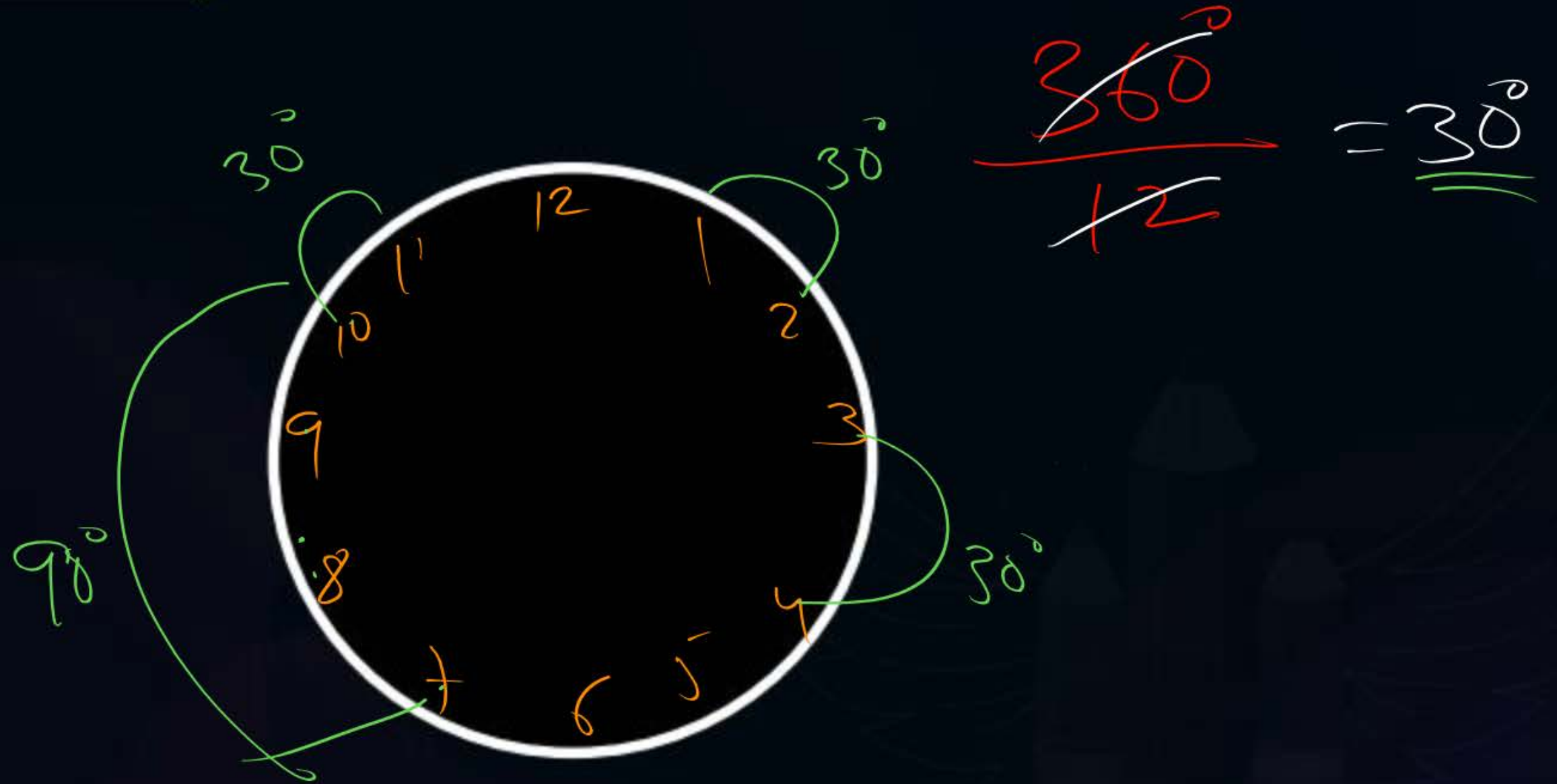
(-)

Minute Hand (M.H.)

Hour Hand (H.H.)

Clockwise

Dial of Clock



Dial of Clock

$$\underline{M.H.} = \frac{360}{8} = 6^\circ/\text{minute}$$

$$\underline{H.H} = \frac{360}{60}$$

$$= \frac{1}{2}^\circ/\text{hour}$$

OR

$$0.5^\circ/\text{hour}$$



To be Noted:

- ✓ Minute Hand Covers 30° in 5 minutes

Time $\rightarrow$ Angle?
Angle $\rightarrow$ Time?

$$30^\circ / 5 =$$

$$6^\circ / \text{min.}$$

- ✓ Hour Hand Covers 30° in 60 minutes

gain or loss

$$30^\circ / 60$$

$$= \frac{1}{2}^\circ / \text{min}$$

or

$$0.5^\circ / \text{min}$$

Relative Speed

$$5.5^\circ / \text{minute}$$

Different Patterns of Questions:

1. Time $\rightarrow$ Angle?
2. Angle $\rightarrow$ Time?
3. Given OP base

5.5 / minute

First Pattern:

4:00

120°

2:00

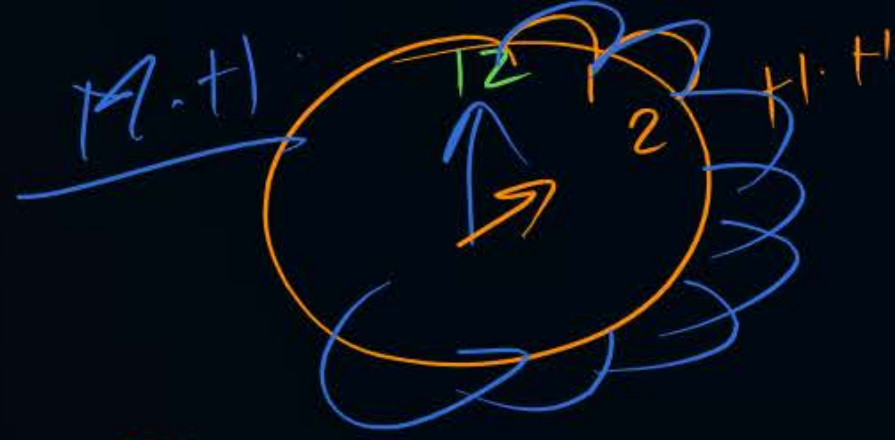
60°

7:00

210°

9:00

270°



Random time given:

$$\begin{array}{l}
 \boxed{S:20} \rightarrow \underline{40^\circ} \rightarrow 320^\circ \\
 S \rightarrow 150^\circ \\
 20 \times 5.5 \Rightarrow 110^\circ \\
 \hline
 40^\circ
 \end{array}$$

Questionnaire:

5:30^w →

15°

345°^w

S → 150°

30 × 5.5 → 165°

15°



Questionnaire:

3:15 → 7.5°

352.5

3 → 90°

15 × 5.5 → 82.5°

7.5°



Questionnaire:

4:40

$\rightarrow 150^\circ$

260°

$4 \rightarrow 120^\circ$

$40 \times 5.5 \Rightarrow 220^\circ$

180°



[MCQ]



#Q. At 9:45, the two hands of a clock make an angle of?

- A** 60°
- B** 45°
- C** $33\frac{1}{3}^\circ$
- D** $22\frac{1}{2}^\circ$

$$9 \rightarrow 270^\circ$$

$$45 \times 5.5 \rightarrow 247.5^\circ$$

$$\underline{\underline{22.5^\circ}}$$

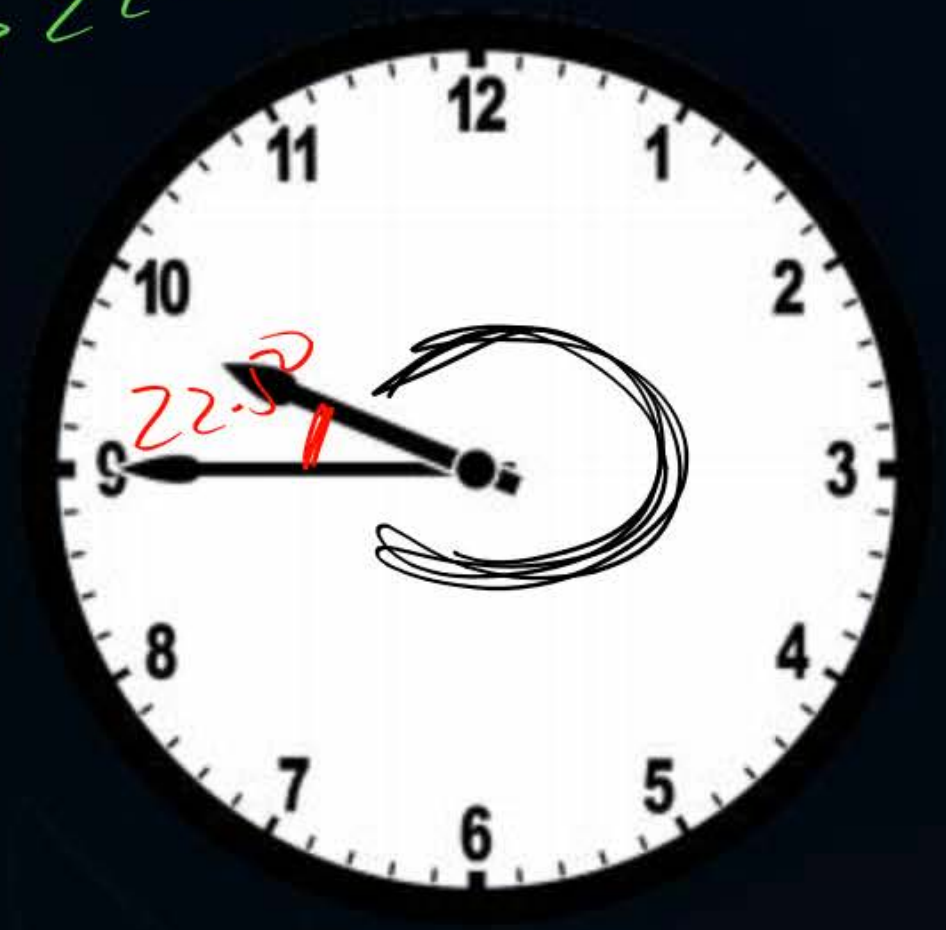
$$337.5^\circ$$

Two Answers:

9:45

337.5°

✓
→ 22.5°



Second Pattern:

Angle $\rightarrow$ Time?





2 mins Summary



Topic

Clock →

1st pattern

Time → Angle?



THANK - YOU